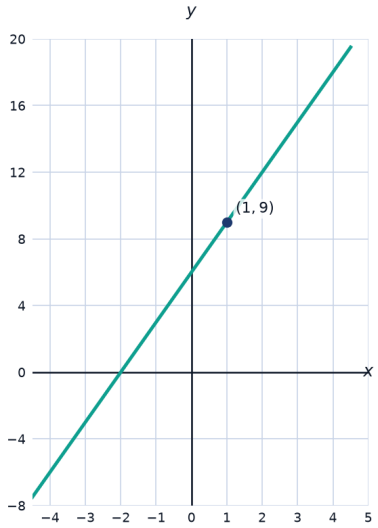
Rational Expressions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

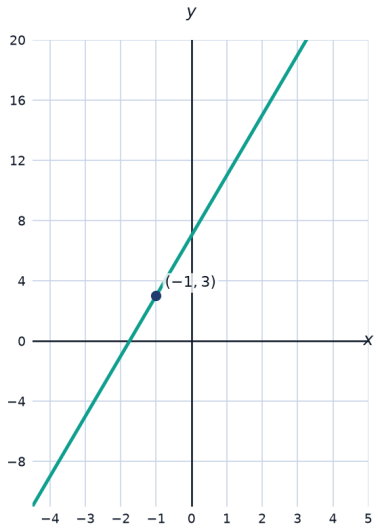
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 7$ in operations with radical expressions, evaluate $f(2)$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

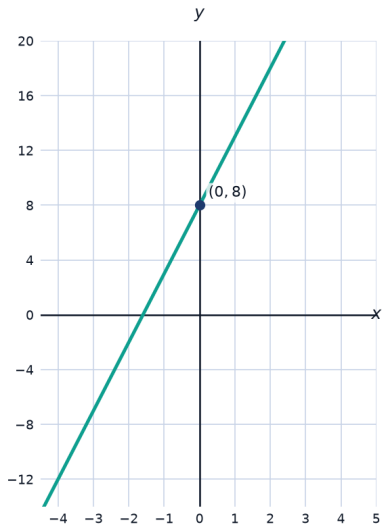
A data set recorded for operations with radical expressions consists of the four values $\{15, 19, 23, 29\}$.

What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 5x + 8$ in graphs and characteristics of square root functions, evaluate $f(2)$.

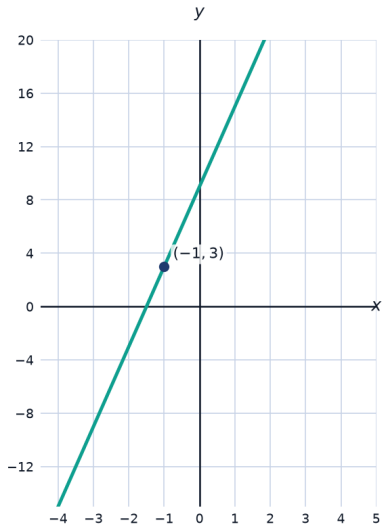


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for graphs and characteristics of square root functions consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 6x + 9$ in graphs and characteristics of cube root functions, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for graphs and characteristics of cube root functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

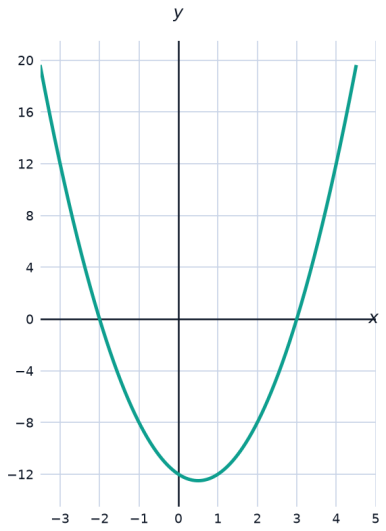
Simplify the complex rational expression: $\frac{\frac{1}{x}-\frac{1}{y}}{\frac{1}{x^2}-\frac{1}{y^2}}$, where $xeqy$ and $xyeq0$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 1 · FoundationsOn Section I Part B of the AP Precalculus exam, a student allocates 50 minutes to answer 12 questions using a graphing calculator. What is the average time per question available in minutes, rounded to the nearest tenth?

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the rational function $r(x) = \frac{2x^2 - 2x - 12}{x^2 - 9}$, find: (a) the domain, (b) the coordinates of any removable discontinuity (hole), (c) the vertical and horizontal asymptotes, and (d) the x- and y-intercepts.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the rational inequality $\frac{x+4}{x-2} \leq 3$ and express the solution set in interval notation.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

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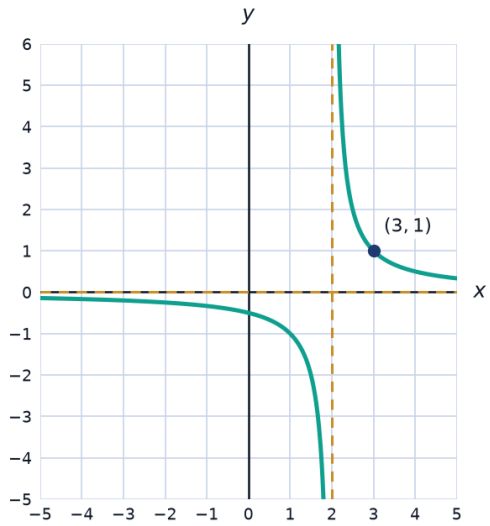
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Trigonometric Functions**

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2 + 5x - 3}{x^2 + 4x + 3}$.

16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Functions & Their Graphs**

Find all real solutions to the rational equation $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2 + x - 6}$.



17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Matrices & Systems**

Decompose the rational expression $R(x) = \frac{7x - 1}{x^2 - x - 6}$ into partial fractions using the Heaviside cover-up method.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Probability & Statistics**

Find the equation of the slant (oblique) asymptote of the rational function $g(x) = \frac{3x^2 - 5x + 7}{x - 2}$, and compute $\lim_{x \to 2^+} g(x)$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Quadratics & Complex Numbers**

Convert the repeating decimal $x = 0.4\overline{36} = 0.4363636\ldots$ into an irreducible rational fraction $\frac{p}{q}$.

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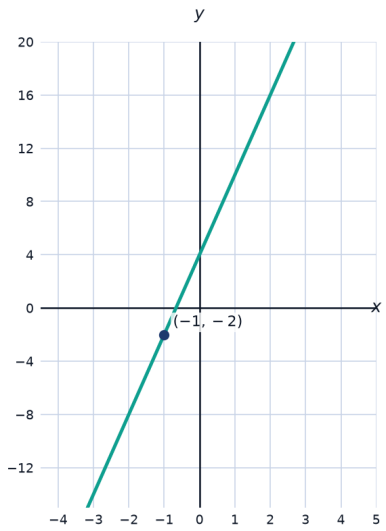
20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Polynomial Functions

Level 3 · AdvancedAn invasive plant covers an area modeled by $A(t) = 120(1.15)^t$ square meters, where t is measured in weeks. Write the equation setup to determine when the area reaches 500 square meters, and find the value of t to three decimal places with correct units.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 4$ in review: laws of exponents, evaluate $f(2)$.



a. 16

b. 22

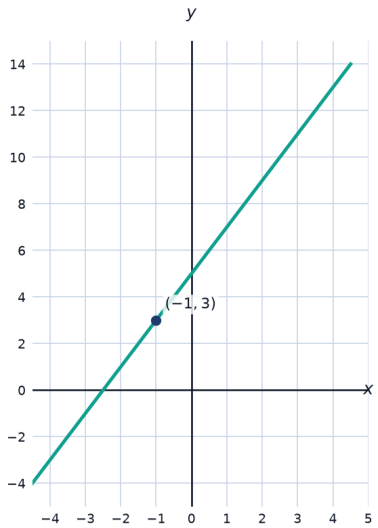
c. 12

d. 20

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 5$ in fractional powers, evaluate $f(2)$.



a. 9

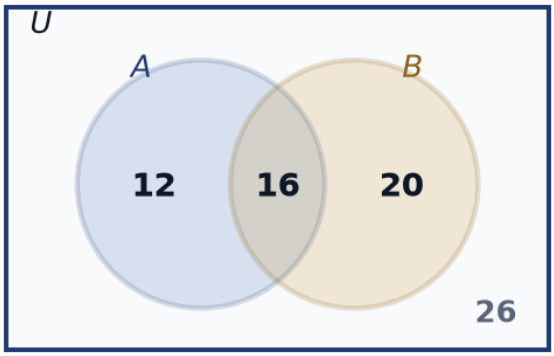
b. 11

c. 4

d. 13

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: laws of exponents consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?



a. 16.5

b. 17.5

c. 18.5

d. 20.5

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for fractional powers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the rational equation $\frac{x+1}{x-3} + \frac{k}{x+1} = \frac{x+13}{x^2-2x-3}$ where k is a constant. What is the sum of all real values of k for which the equation has no real solution?

a. -10

b. -7

c. -3

d. 10

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of the rational function $g(x) = \frac{ax^2+bx+1}{x^2-4}$ has a horizontal asymptote at $y = 2$ and a vertical asymptote at $x = 2$. Given that $g(0) = -1/4$, find the value of a.

a. 1

b. 2

c. 3

d. 4

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the rational function $f(x) = \frac{x^2-2x+c}{x^2+2x+c}$ where c is a real constant. If the domain of f is the set of all real numbers and the maximum value of $f(x)$ is 4, what is the value of c?

a. $\frac{25}{9}$

b. $\frac{9}{4}$

c. 3

d. $\frac{16}{5}$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which of the following is equivalent to the expression $\frac{x^2-25}{2x-10}$ for all $xeq5$?

a. $\frac{x+5}{2}$

b. $\frac{x-5}{2}$

c. $x+5$

d. $\frac{x+5}{2(x-5)}$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Which of the following expressions is equivalent to $\frac{x^2-9}{x-3}$ for all values of x where the expression is defined?

a. $x+3$

b. $x-3$

c. x^2-3

d. $x+9$

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Find the sum $\frac{1}{x(x+1)}+\frac{1}{(x+1)(x+2)}+\frac{1}{(x+2)(x+3)}$ using partial fractions.

a. $\frac{1}{x+3}$

b. $\frac{3}{(x+1)(x+2)}$

c. $\frac{3}{x(x+3)}$

d. $\frac{1}{x}-\frac{1}{x+3}$

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Write $\frac{x}{(x+1)^2}$ in partial fractions.

a. $\frac{1}{x+1}-\frac{1}{(x+1)^2}$

b. $\frac{x}{(x+1)^2}$ cannot be decomposed

c. $\frac{1}{x+1}+\frac{1}{(x+1)^2}$

d. $\frac{1}{(x+1)^2}$

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Determine if $f(x)=\frac{3x^2-5}{x^2+2}$ has any vertical asymptotes.

a. Yes at $x=0$

b. Yes at $x=\pm\sqrt{2}$

c. Yes at $y=3$

d. No (denominator $x^2+2>0$ always)

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

What happens to $f(x)=\frac{1}{x}$ as $x\to 0^+$?

a. $f\to 0$

b. $f\to -\infty$

c. $f\to +\infty$

d. $f\to 1$

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which of these is a rational function?

a. $f(x) = \sqrt{x}$

b. $f(x) = \frac{x^2 - 1}{x + 3}$

c. $f(x) = x^{2/3}$

d. $f(x) = 2^x$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find all solutions of $\frac{x^2 + 2x}{x^2 - 1} = 1 + \frac{3}{x - 1}$.

a. $x = 1$

b. No solution

c. $x = -4$

d. $x = 4$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the partial fraction decomposition of $\frac{1}{x^3 - x}$.

a. $\frac{1}{x(x-1)} - \frac{1}{x+1}$

b. $-\frac{1}{x} + \frac{1/2}{x-1} + \frac{1/2}{x+1}$

c. $\frac{1}{x} + \frac{1}{x-1} + \frac{1}{x+1}$

d. $\frac{1}{x^2 - 1} - \frac{1}{x}$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the exact solution of $\frac{x}{x-1} + \frac{2}{x+1} = \frac{6}{x^2 - 1}$.

a. $x = -2$

b. No solution

c. $x = 3$

d. $x = 2$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Probability & Statistics

Is $f(x) = \frac{1}{x^2 + 4}$ always positive?

a. No

b. Only for $x > 0$

c. Only for $x < 0$

d. Yes

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the vertical asymptotes of $f(x) = \frac{x + 1}{x^2 - 4}$.

a. $x = \pm 2$

b. No VA

c. $x = -1, \pm 2$

d. $x = 2$ only

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Polynomial Functions

Find the x-intercepts and vertical asymptotes of $f(x) = \frac{x^2 - x - 6}{x^2 - 4}$.

a. x-int: $x = 3, -2$ (but $x = -2$ is a hole); VA: $x = 2$

b. x-int: $x = 3$; VA: $x = \pm 2$

c. x-int: $x = 3, -2$; VA: $x = \pm 2$

d. x-int: $x = 3$; VA: $x = 2$ only

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the x-intercept of $f(x) = \frac{x - 3}{x + 2}$.

a. $x = -2$

b. $x = 2$

c. $x = 3$

d. $x = -3$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Radicals & Rational Exponents

The domain of $f(x) = \frac{1}{x - 3}$ excludes:

a. $x = 3$

b. $x = 0$

c. All reals

d. $x = -3$

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Sequences & Series

Find the vertical asymptote of $f(x) = \frac{2}{x+5}$.

a. $x = -5$

b. $y = -5$

c. $x = 5$

d. $x = 0$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Conic Sections

Find the inverse of $f(x) = \frac{3x+2}{x-1}$ and state its domain.

a. $f^{-1}(x) = \frac{x+2}{x-3}$; domain: all reals except $x = 3$

b. $f^{-1}(x) = \frac{3x-2}{x+1}$

c. $f^{-1}(x) = \frac{x+2}{x+3}$

d. $f^{-1}(x) = \frac{x-2}{x+3}$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Trigonometric Functions

Identify: $\frac{p(x)}{q(x)}$ is called a _____ function.

a. Irrational

b. Rational

c. Polynomial

d. Transcendental

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